

Kent Christie

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Role: HSE / Environmental & Safety Advisor — Metal Recovery & Hazardous Waste

Results-driven, fully bilingual (Eng/Span) Health, Safety & Environment leader with 25+ years of experience in design, construction, and operations for the mining and manufacturing industries, in numerous countries, including Saudi Arabia, Mauritania, Ghana, Australia, Canada, USA, UK, Chile, and Mexico, among others. Develops and implements management systems patterned on ISO 45001 and ISO 14001. Committed to a “continual improvement”

CAREER ACCOMPLISHMENTS

Project Hazard Identification and Assessment: Experienced leader in project hazard analysis, using tools such as FMEA (Failure Modes & Effects Analysis), HAZOP (Hazard & Operability Study), PHA (Process Hazard Analysis), RCA (Root Cause Analysis such as TapRoot, Ishikawa, Pareto, Fault Tree Analysis, and RAMS (Risk Assessments & Method Statements). Projects included a copper recovery/recycling plant; wastewater treatment plant; the addition of a baghouse filtration system. CIL plant (Carbon-In-Leach); crushing and grinding upgrade; electrowinning circuit; ore transport belt system; expanded tailings facility; solid/liquid separation, flotation and thickening; Developed controls with team members and verified implementation of action items. Ensured that HSE reviews and assessments were integrated into all phases of the project, i.e. Design, Construction, Commissioning and Operations.

Regulatory Compliance: Compliance-related citations and fines decreased between 19% and 42% at Capstone, Excelsior Mining, Kinross, and Barrick. Drove the development and successful implementation of a regulatory and internal standards compliance program as part of the HSEMS. This included wastewater treatment and discharge programs; hazardous and special waste handling and disposal; and air emissions. Developed and implemented regulatory compliance programs according to the governing agencies, including MHRS (Saudi Arabia); MoL (Mauritania); DFI (Ghana); MSHA, OSHA, NFPA (USA); SERNAGEOMIN (Chile); STPS (Mexico), MTE (Brazil); OSINERGMIN (Peru); and ARCOM (Ecuador).

Health & Safety Management Systems/Policies/Standards: 32% reduction in Lost-Time Incidents and a 34% reduction in Total Recordable Injuries at Capstone Copper, and zero (0) Lost-Time Incidents at Kinross Ecuador, Excelsior Mining, Shell Gabon, and First Mode during my tenure. This was accomplished by directing the implementation and continual improvement of risk-based HSE management systems, including ISO 45001, the International Cyanide Management Code, the Copper Mark and ISO 14001 at Kinross Gold (Ecuador, Chile, Mauritania), Capstone Copper (Chile, Mexico, USA), Excelsior Mining (USA), Drill Corp (Saudi Arabia) and First Mode (Chile, Brazil, USA, Australia), and Shell Gabon.

Accident and Near Miss Investigations: Zero (0) fatalities or serious injuries from 2021 to present at the Capstone Cozamin underground mine (Mexico). Operation averaged one (1) fatality every two (2) years since acquisition. Implemented an improved accident investigation process (ICAM), Root Cause Analysis methodology, and CAPA process (Corrective Action and Preventive Action). Led selected investigations of serious injury and fatalities at Barrick Gold's underground mine (USA-Turquoise Ridge); Kinross' underground mine (Ecuador-Fruta del Norte); Excelsior Mining's in situ copper mine (USA-Johnson Camp); Capstone's underground mine (Mexico-Cozamin), as well as open pit operations in Chile and the USA. Led team to develop or improve investigation processes, including methodologies, root-cause analysis techniques, training, action planning and follow-up.

PROFESSIONAL EXPERIENCE

HS&E Program Assessment/Development

KC International, Owner Saudi Arabia, USA, Mexico, Nicaragua

2024-present

- Conducted assessments of H&S Management Systems and programs for an exploration drilling services company in Saudi Arabia working for the state mining company, Ma'aden. Included management system roadmap development,

critical risk management, training, regulatory compliance, and action monitoring/verification. The company achieved ISO 45001 certification using the improvements that were recommended.

- Provided H&S strategy and program development support for mine water treatment facilities at the Hermosa mine in southern Arizona, USA.
- Working as an expatriate contractor to a gold mining company in Nicaragua, developing and implementing HSE assessment and communication programs, focused on all leadership levels within the company. Included the El Limon, La Libertad and EBM mine sites. Contributed to Lost Time Injury rate of 1.70, representing a reduction of 23%, as well as 0 operational site fatalities.

Director, Environmental, Health and Safety

First Mode Seattle, Washington USA

2024

- Led the HSE assessment and controls implementation program for research, design, testing and manufacturing of hydrogen fuel cells and hybrid batteries used in mining haul trucks. This included PHAs and FMEAs incorporated in each phase of the project.
- Directed a team of 4 professionals in developing an innovative, company-wide HSE Management System applicable to the U.S., Chile, Brazil, United Kingdom, and Australia. This included Enterprise Risk Management policies, procedures, programs and tools. Targeted ISO 45001 and 14001 certifications in the near term.
- Responsible for ensuring HSE regulatory compliance with MSHA, OSHA, EPA, NFPA, MTE/MMA (Brazil), SERNAGEOMIN (Chile), and WorkSafe (Western Australia). Worked jointly with MSHA and L&I (Washington) to improve safety.
- Developed and delivered HSE-related training courses to all levels of the company.

Director, Health, Safety and Environment

Capstone Copper Corporation Phoenix, Arizona USA

2020-2023

- Directed HAZOPs and Risk Assessments of a copper metal recovery/recycling plant at the Pinto Valley mine in Arizona. Also collaborated in the development of Safe Work Procedures for the operation of the facility.
- Oversaw hazard waste disposal programs
- Mentored Mexico's Cozamín underground operational and HSE staff in achieving compliance with STPS (NOM-023-STPS-2003), Protección Civil, SEMARNAT/PROFEPA, IMMS and ISSSTE requirements. The operation achieved approval under the PASST, ELSSA, Industria Limpia and ESR programs, as well as certification under the international standard, The Copper Mark. Modifications of the HSE programs at Cozamín resulted in 0 fatalities during the last 5 years. Previous fatalities averaged one every two years.
- Initiated a Critical Controls management program to address serious injuries and fatalities (SIFs).
- Implemented the Incident Cause Analysis Method (ICAM) at all 4 operating sites and 1 project (Arizona, Mexico, Chile)
- Led the development of company-wide management and technical standards for Health and Safety, based upon industry-recognized Best Management Practice. Part of the HSE Management System.

EHSS Manager (contractor)

KC International Consulting, Owner Cochise Mining District, Arizona USA

2019-2020

- Collaborated in an engineering design review and HAZOP assessment of hazards for a refitted copper electrowinning circuit at the Gunnison project in Arizona, USA.
- Implemented a hazard assessment program resulting in multiple safety improvements across the operation.
- Initiated management system development patterned after ISO 45001, including critical risk control program.
- Implemented a data analysis and problem-solving approach to identify areas of improvement.
- Developed a 3-year strategic plan and budget for EHSS.

Director, Safety & Sustainability

Kinross Gold Corporation Denver, Colorado USA

2011-2018

- Led the Engineered Projects Risk Assessment program (FMEAs, HAZOPs) at all sites, including highly complex assessment at the Tasiast mine in Mauritania.

- Led the corporate participation in the design and selection of wastewater treatment facilities at the Fruta del Norte mine in Ecuador.
- Directed certification/re-certification efforts at operating sites in compliance with the International Cyanide Management Code, successfully achieving certification at all operations with 0 incidents.
- Responsible for multiple company-wide programs at 11 underground and surface operating sites, projects and exploration activities located in Mexico, Chile, Ecuador, Brazil, Russia, Ghana, Mauritania and the USA.
- Directed a risk-based audit program including Health & Safety, Environment and Community Relations, which identified root causes, and corrective/preventive actions. Tracked completion and effectiveness of all actions. The audit program contributed to a reduction in the Lost Time Injury Frequency rate of 38% and zero (0) reportable releases to the environment.

Director, HSE Audits

Barrick Gold Corporation Salt Lake City, Utah USA

2006-2010

Managed 5-person, full-time team, auditing 28 underground and surface operations across 4 continents for a multi-billion-dollar gold mining concern. Auditing focus included health & safety leadership, environmental, legal compliance, management systems, and data management.

- Led or provided oversight for more than 50 EHS audits in North America, South America, Africa, and Australia, contributing to a reduction in Lost Time Injuries (25%) and reportable releases (35%).
- Directed the development and usage of audit program protocols, checklists, and toolbox. This permitted individual operations to conduct their own internal audits.
- Established a user-friendly corrective action tracking and reporting tool, including root cause analysis, trend analysis and reporting to corporate management.
- Trained more than 40 internal auditors enabling mining operations to conduct their own internal audits.

EDUCATION, LANGUAGES, CERTIFICATION and TRAINING

B.S. Science, Economics – University of Utah

B.A. Arts, Spanish – University of Utah

Certificate, International Relations – University of Utah

English – Native or bilingual proficiency

Spanish – Native or bilingual proficiency

Certifications

CMSP 2025 (Certified Mine Safety Professional)

HAZWOPER 2025 (Hazardous Waste Operations and Emergency Response)

MSHA Authorized Mine Safety Instructor 2024 (Blue Card)

First Aid/CPR/AED 2026 (Red Cross)

Mental First Aid 2023

ISO 14001 Lead Auditor (requires renewal)

International Cyanide Management Code approved auditor (requires renewal)

OSHA 30-hr General Industry

Training Courses

Project Manager in Action, FMEA (Failure Modes & Effects Analysis) Risk Assessment, HAZWOPER 40 HR, ICAM (Incident Cause Analysis Method), OSHA 45001 Lead Auditor, ISO 45001, OHSAS 18001, ISO 140001, International Cyanide Management Code, MSHA Regulations, OSHA 30 HR, TapRoot Incident Investigation and Root Cause Analysis, Confined Space, Lockout Tagout, RCA (Root Cause Analysis), NOM-023-STPS Mexican H&S regulations, Compatibility/Handling/Storage of Hazardous Materials, Working at Heights, MSHA New Miner (Underground and Surface), SAFEmap Safety Leadership,